

Resonant Inelastic X-ray Scattering by Dynamical Mean-Field Theories

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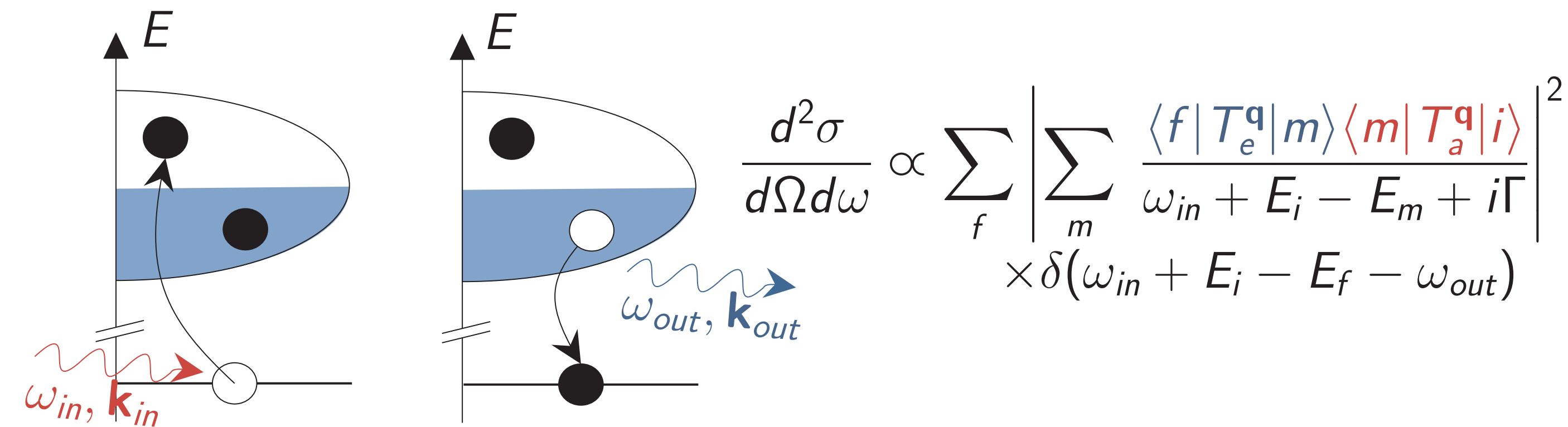
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Resonant Inelastic X-ray Scattering

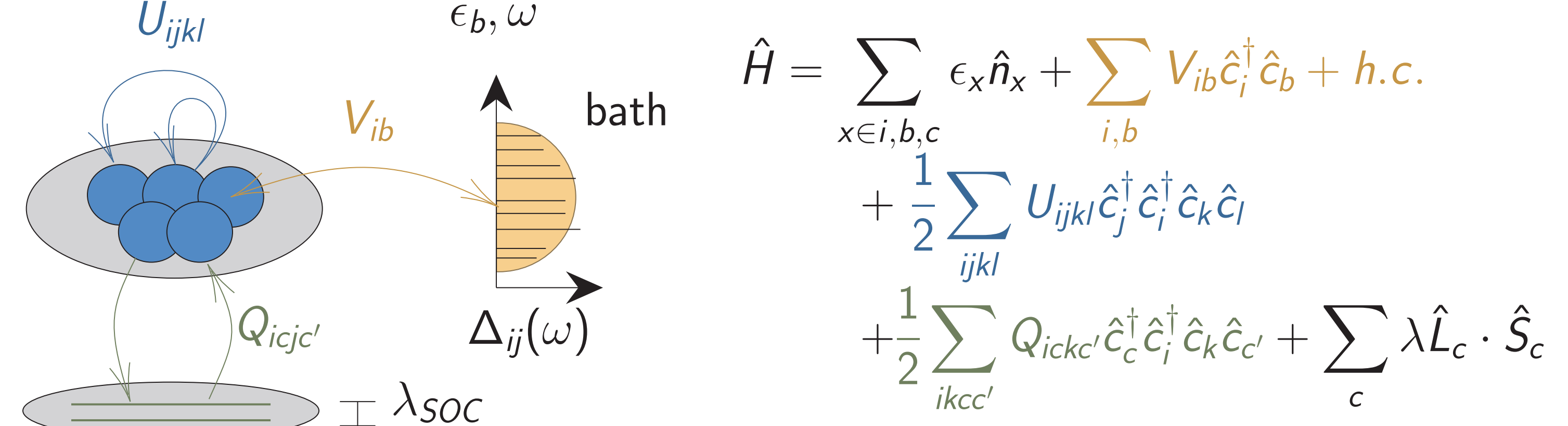
Resonant Inelastic X-ray Scattering (RIXS) is a two process spectroscopy technique with core absorption and emission. It allows to probe a lot of different excitations (phonons, magnons, d - d , charge transfer).



The quantity of main interest is the cross section σ , where the groundstate ($|i\rangle, E_i$) appears, as well as the intermediate absorption state ($|m\rangle = T_a^q |i\rangle, E_m$) and the final emission state ($|f\rangle = T_e^q |m\rangle, E_f$). Γ is the quasiparticle lifetime.

Augmented Anderson Impurity Model (AAIM)

The model of interest is the AAIM, corresponding to an impurity with core and valence states, the latter hybridizing with a non-interacting bath.



Building the model

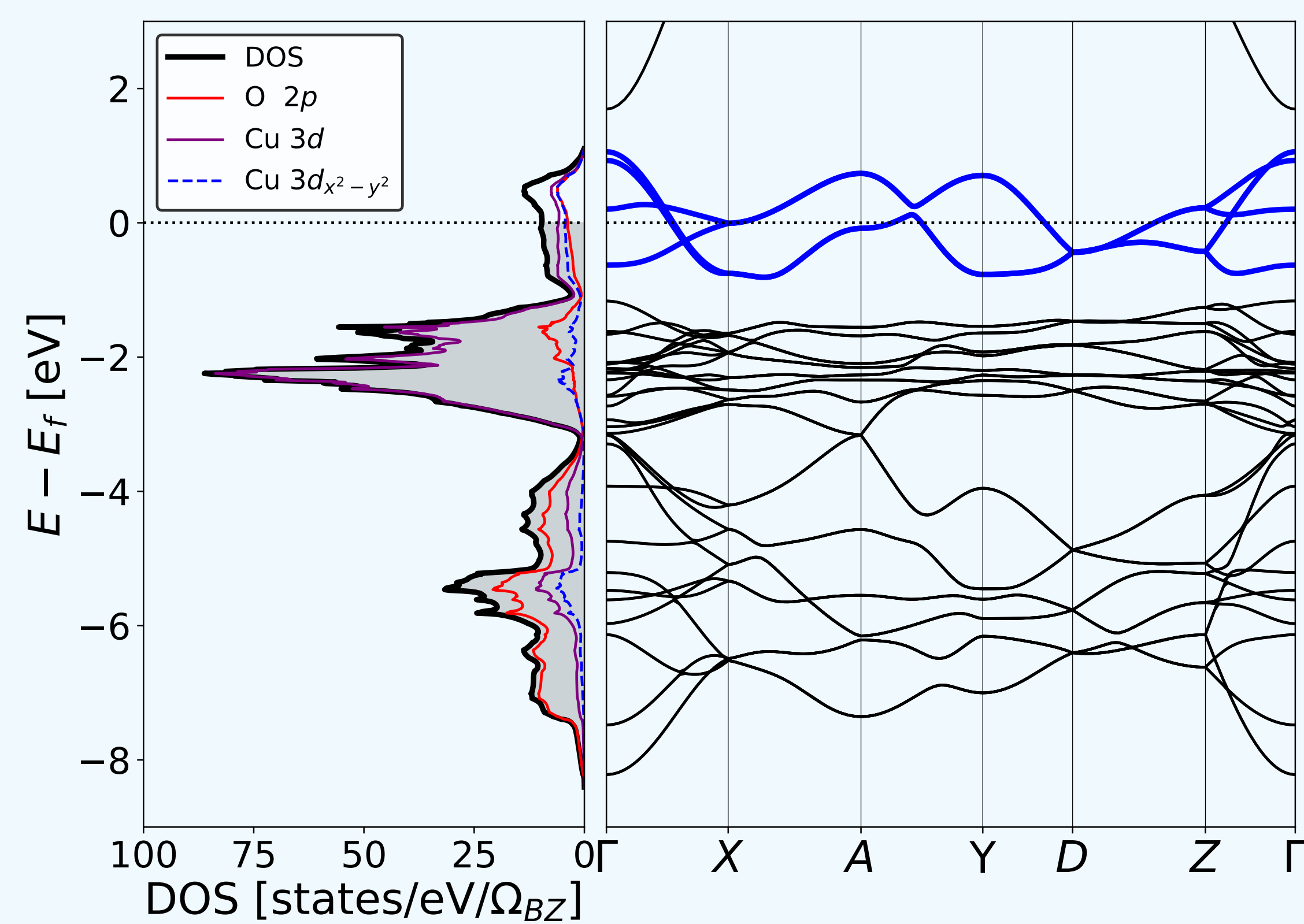
- Density Functional Theory
- Dynamical Mean Field Theory

Solving the model

- Matrix Product State formalism
- Exact Diagonalization (Lanczos)

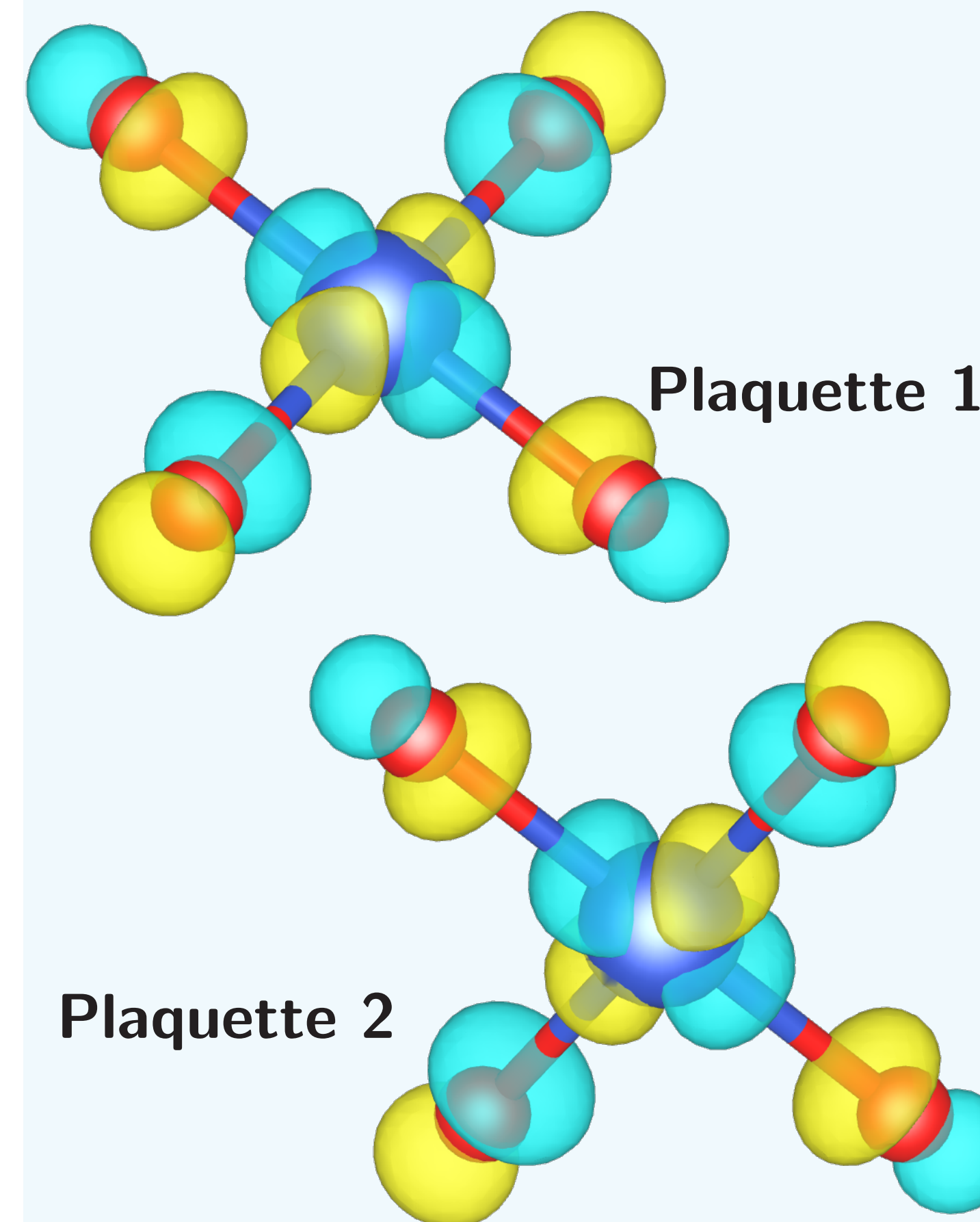
Density Functional Theory (DFT)

We do a one band model benchmark with copper oxide CuO



Wannier Functions (WF)

WF are Fourier transform of BS: $|\mathbf{R}\mathbf{n}\rangle = \int_{\text{BZ}} d\mathbf{k} e^{-i\mathbf{k}\mathbf{R}} \sum_m U_{mn}^{(\mathbf{k})} |\phi_{n,\mathbf{k}}\rangle$



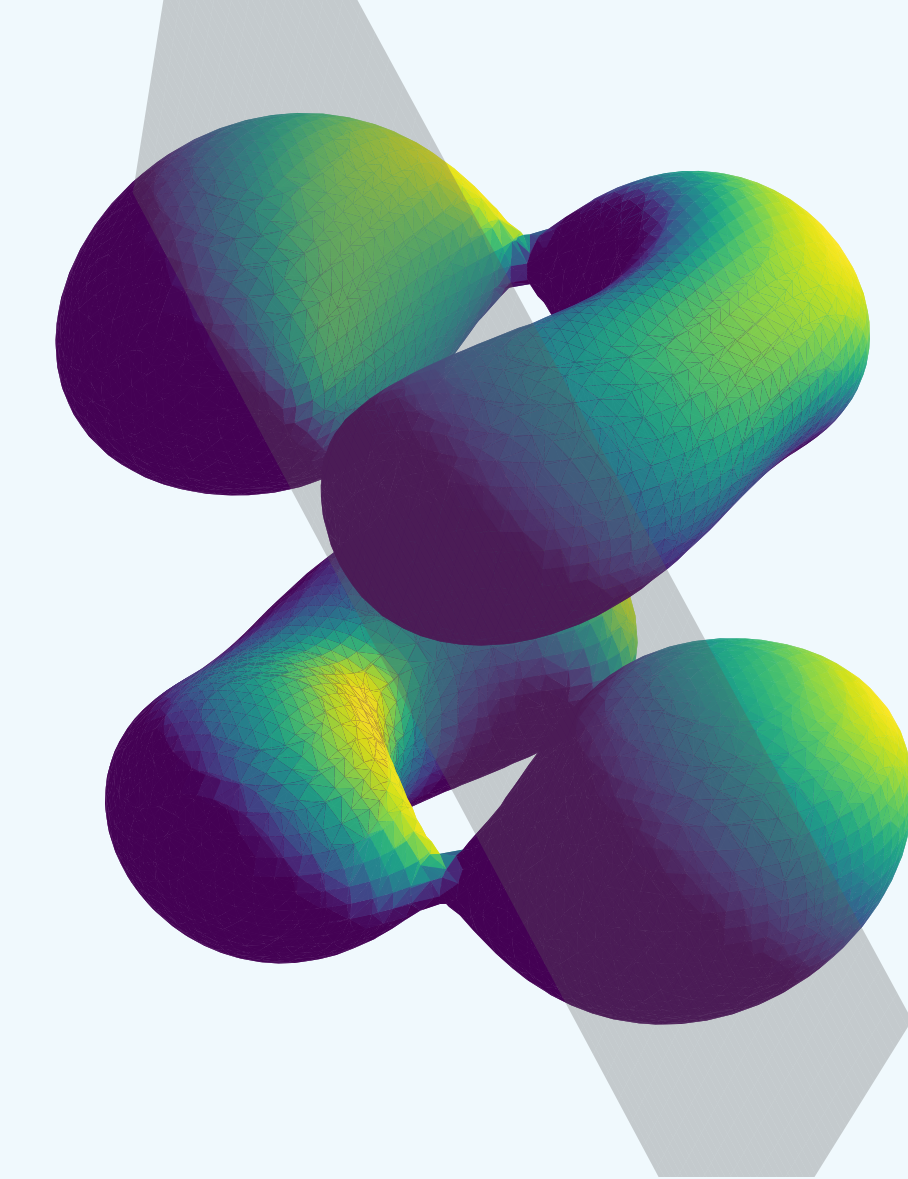
- We maximally localize them with respect to the $U^{(\mathbf{k})}$ gauge.
- These WF are localized Cu-3d_{x²-y²} orbitals centered on each Cu site $\mathbf{R}$, hybridized with nearest O-2p orbitals.
- The 3s orbital is a core state (see DOS), we use an atomic wavefunction.

Numerical Results and Comparison with Experimental Data

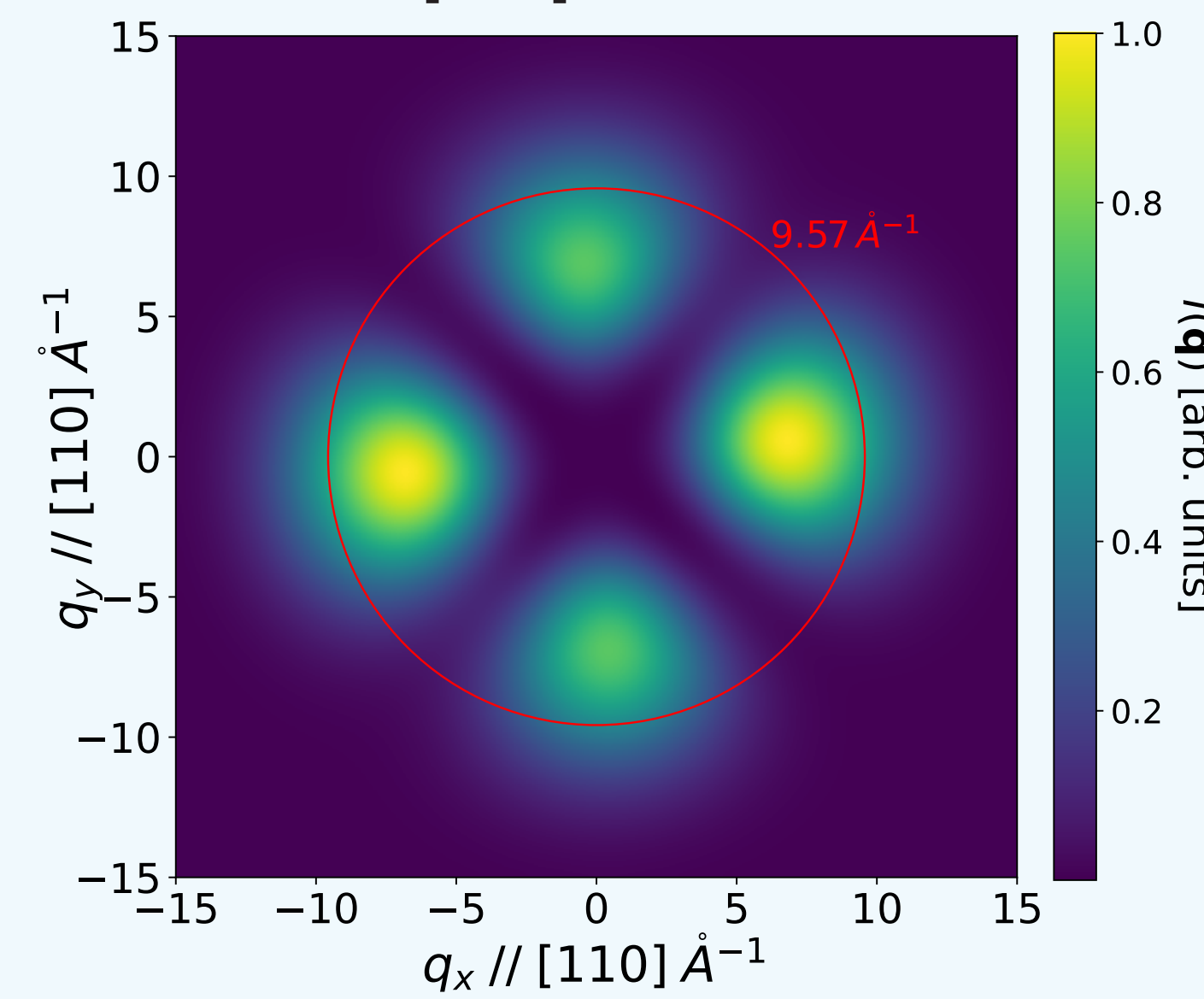
- Two inequivalent copper atoms
 - One 3d_{x²-y²} hole on each ineq. copper atom
- $$\rightarrow I(\mathbf{q}) = \sum_i |\langle 3d_{x^2-y^2}^{(i)} | e^{i\mathbf{q}\mathbf{r}} | 3s^{(i)} \rangle|^2$$

- **Left:** non-orthogonal plaquettes $\rightarrow$ mixing of NRIXS contributions from inequivalent atoms.
- **Middle:** probe of the [110] plane, nearly parallel to one plaquette. Asymmetry between q_x and q_y .
- **Right:** polar plot compared with experiment (red and black dots) [2], showing good agreement. We observe a small contribution from the second plaquette.

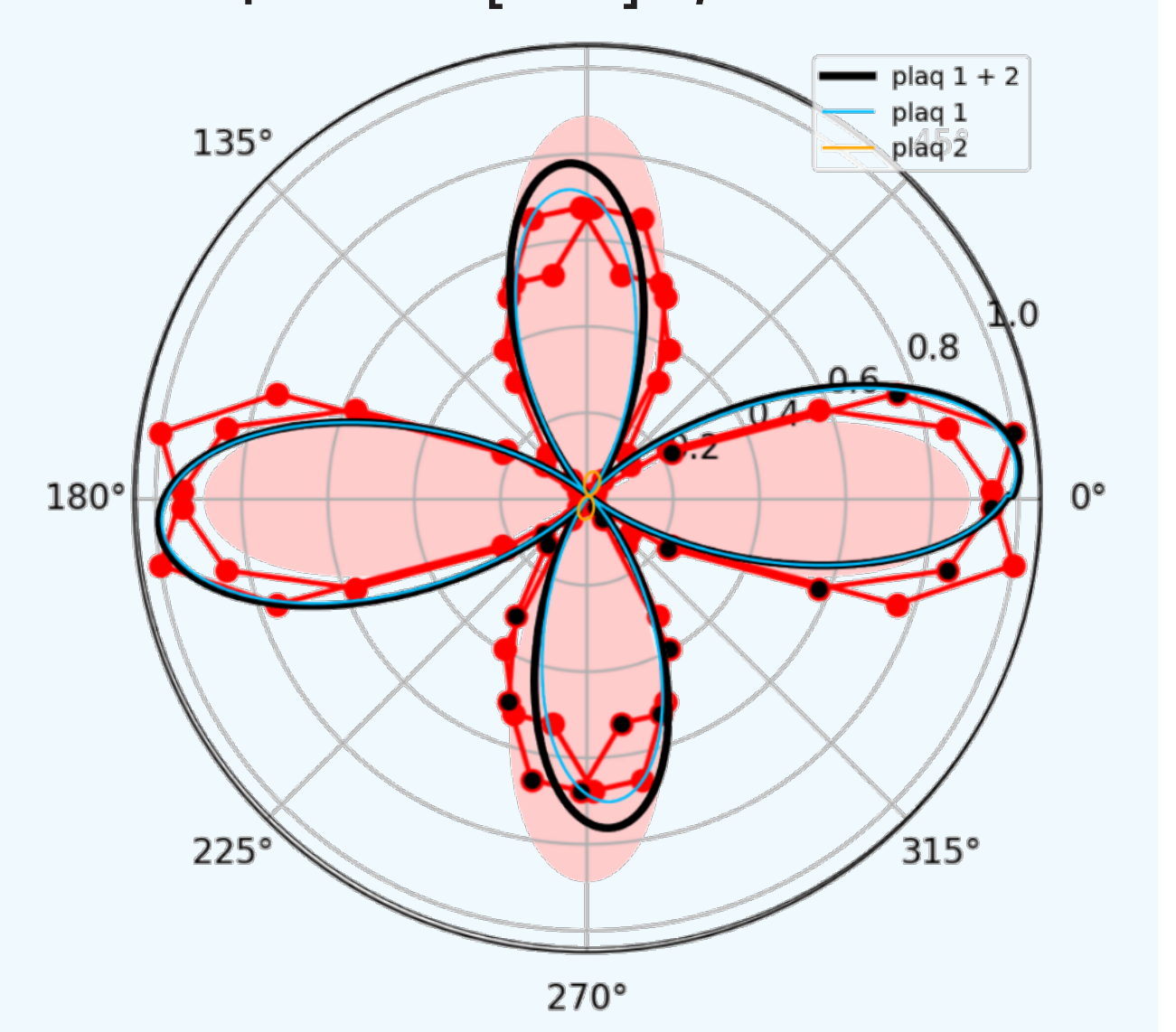
Full angular dependence



[110] plane



Polar plot of [110] $q = 9.57 \text{ \AA}^{-1}$



Where does the Asymmetry come from ?

Since the second plaquette contributes only weakly to the previous figure, the asymmetry mainly originates from the fact that the probed plane is not exactly aligned with the plaquette plane. Does a plane parallel to a plaquette still exhibit such an asymmetry?

We present calculations for a plane parallel to one plaquette, with $q = 9.57 \text{ \AA}^{-1}$.

- The second plaquette now contributes significantly to the overall asymmetry.
- It slightly tilts the angular distribution.
- A residual asymmetry remains in the first plaquette due to inequivalent Cu-O bonds.

Conclusion and Outlook

We developed a first-principles framework combining DFT+U and Wannier functions to compute the angular dependence of NRIXS spectra. Applied to CuO, the method reproduces the main experimental features and identifies the microscopic origin of the observed asymmetry.

Outlook:

- Future work will focus on correlated materials requiring the explicit inclusion of the density matrix in Eq. (1). We welcome suggestions of candidate systems.
- Fully self-consistent DFT+DMFT calculations also constitute also a promising extension, allowing electronic correlations to enter the Wannier functions via the feedback loop.

References

[1] H. Yavas, M. Sundermann, K. Chen, A. Amorese, A. Severing, H. Gretarsson, M. W. Haverkort, and L. H. Tjeng. *Direct imaging of orbitals in quantum materials*. *Nature Physics*, 15(6) :559–562, 2019.

[2] Right figure of the Motivation box : experimental results of orbital imaging of copper oxide at 300 K by Victor Roland and Jean-Pascal Rueff, with whom we are collaborating. The NRIXS spectra comes from paper [1] for NiO.